

Python Module Index

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[_](#)

[__future__](#) *Future statement definitions*

[__main__](#) *The environment where top-level code is run. Covers command-line interfaces, import-time behavior, and ``\_\_name\_\_ == '\_\_main\_\_'``.*

[_thread](#) *Low-level threading API.*

[_tkinter](#) *A binary module that contains the low-level interface to Tcl/Tk.*

a

[abc](#) *Abstract base classes according to :pep:3119`.*

[aifc](#) **Deprecated:** *Removed in 3.13.*

[annotationlib](#) *Functionality for introspecting annotations*

[argparse](#) *Command-line option and argument parsing library.*

[array](#) *Space efficient arrays of uniformly typed numeric values.*

[ast](#) *Abstract Syntax Tree classes and manipulation.*

[asynchat](#) **Deprecated:** *Removed in 3.12.*

[asyncio](#) *Asynchronous I/O.*

[asyncore](#) **Deprecated:** *Removed in 3.12.*

[atexit](#) *Register and execute cleanup functions.*

[audioop](#) **Deprecated:** *Removed in 3.13.*

b

[base64](#) *RFC 4648: Base16, Base32, Base64 Data Encodings; Base85 and Ascii85*

[bdb](#) *Debugger framework.*

[binascii](#) *Tools for converting between binary and various ASCII-encoded binary representations.*

[bisect](#) *Array bisection algorithms for binary searching.*

[builtins](#) *The module that provides the built-in namespace.*

[bz2](#) *Interfaces for bzip2 compression and decompression.*

c

[calendar](#) *Functions for working with calendars, including some emulation of the Unix cal program.*

[cgi](#) **Deprecated:** *Removed in 3.13.*

[cgitb](#) **Deprecated:** *Removed in 3.13.*

chunk	Deprecated: <i>Removed in 3.13.</i>
cmath	<i>Mathematical functions for complex numbers.</i>
cmd	<i>Build line-oriented command interpreters.</i>
code	<i>Facilities to implement read-eval-print loops.</i>
codecs	<i>Encode and decode data and streams.</i>
codeop	<i>Compile (possibly incomplete) Python code.</i>
+ collections	<i>Container datatypes</i>
colorsys	<i>Conversion functions between RGB and other color systems.</i>
compileall	<i>Tools for byte-compiling all Python source files in a directory tree.</i>
+ compression	
+ concurrent	
configparser	<i>Configuration file parser.</i>
contextlib	<i>Utilities for with-statement contexts.</i>
contextvars	<i>Context Variables</i>
copy	<i>Shallow and deep copy operations.</i>
copyreg	<i>Register pickle support functions.</i>
cProfile	
crypt	Deprecated: <i>Removed in 3.13.</i>
csv	<i>Write and read tabular data to and from delimited files.</i>
ctypes	<i>A foreign function library for Python.</i>
+ curses	<i>An interface to the curses library, providing portable terminal handling.</i>
d	
dataclasses	<i>Generate special methods on user-defined classes.</i>
datetime	<i>Basic date and time types.</i>
+ dbm	<i>Interfaces to various Unix "database" formats.</i>
decimal	<i>Implementation of the General Decimal Arithmetic Specification.</i>
difflib	<i>Helpers for computing differences between objects.</i>
dis	<i>Disassembler for Python bytecode.</i>
distutils	Deprecated: <i>Removed in 3.12.</i>
doctest	<i>Test pieces of code within docstrings.</i>
e	
+ email	<i>Package supporting the parsing, manipulating, and generating email messages.</i>
+ encodings	<i>Encodings package</i>
ensurepip	<i>Bootstrapping the "pip" installer into an existing Python installation or virtual environment.</i>
enum	<i>Implementation of an enumeration class.</i>
errno	<i>Standard errno system symbols.</i>

f

faulthandler	<i>Dump the Python traceback.</i>
fcntl	<i>The <code>fcntl()</code> and <code>ioctl()</code> system calls.</i>
filecmp	<i>Compare files efficiently.</i>
fileinput	<i>Loop over standard input or a list of files.</i>
fnmatch	<i>Unix shell style filename pattern matching.</i>
fractions	<i>Rational numbers.</i>
ftplib	<i>FTP protocol client (requires sockets).</i>
functools	<i>Higher-order functions and operations on callable objects.</i>

g

gc	<i>Interface to the cycle-detecting garbage collector.</i>
getopt	<i>Portable parser for command line options; support both short and long option names.</i>
getpass	<i>Portable reading of passwords and retrieval of the userid.</i>
gettext	<i>Multilingual internationalization services.</i>
glob	<i>Unix shell style pathname pattern expansion.</i>
graphlib	<i>Functionality to operate with graph-like structures</i>
grp	<i>The group database (<code>getgrnam()</code> and friends).</i>
gzip	<i>Interfaces for gzip compression and decompression using file objects.</i>

h

hashlib	<i>Secure hash and message digest algorithms.</i>
heapq	<i>Heap queue algorithm (a.k.a. priority queue).</i>
hmac	<i>Keyed-Hashing for Message Authentication (HMAC) implementation</i>
html	<i>Helpers for manipulating HTML.</i>
http	<i>HTTP status codes and messages</i>

i

idlelib	<i>Implementation package for the IDLE shell/editor.</i>
imaplib	<i>IMAP4 protocol client (requires sockets).</i>
imghdr	Deprecated: Removed in 3.13.
imp	Deprecated: Removed in 3.12.
importlib	<i>The implementation of the import machinery.</i>
inspect	<i>Extract information and source code from live objects.</i>
io	<i>Core tools for working with streams.</i>
ipaddress	<i>IPv4/IPv6 manipulation library.</i>
itertools	<i>Functions creating iterators for efficient looping.</i>

j

 [json](#) *Encode and decode the JSON format.*

k

[keyword](#) *Test whether a string is a keyword in Python.*

l

[linecache](#) *Provides random access to individual lines from text files.*

[locale](#) *Internationalization services.*

 [logging](#) *Flexible event logging system for applications.*

[lzma](#) *A Python wrapper for the liblzma compression library.*

m

[mailbox](#) *Manipulate mailboxes in various formats*

[mailcap](#) **Deprecated:** *Removed in 3.13.*

[marshal](#) *Convert Python objects to streams of bytes and back (with different constraints).*

[math](#) *Mathematical functions (sin() etc.).*

[mimetypes](#) *Mapping of filename extensions to MIME types.*

[mmap](#) *Interface to memory-mapped files for Unix and Windows.*

[modulefinder](#) *Find modules used by a script.*

[msilib](#) **Deprecated:** *Removed in 3.13.*

[msvcrt](#) *Miscellaneous useful routines from the MS VC++ runtime.*

 [multiprocessing](#) *Process-based parallelism.*

n

[netrc](#) *Loading of .netrc files.*

[nis](#) **Deprecated:** *Removed in 3.13.*

[nntplib](#) **Deprecated:** *Removed in 3.13.*

[numbers](#) *Numeric abstract base classes (Complex, Real, Integral, etc.).*

o

[operator](#) *Functions corresponding to the standard operators.*

[optparse](#) *Command-line option parsing library.*

 [os](#) *Miscellaneous operating system interfaces.*

[ossaudiodev](#) **Deprecated:** *Removed in 3.13.*

p

 [pathlib](#) *Object-oriented filesystem paths*

[pdb](#) *The Python debugger for interactive interpreters.*

[pickle](#) *Convert Python objects to streams of bytes and back.*

pickletools	<i>Contains extensive comments about the pickle protocols and pickle-machine opcodes, as well as some useful functions.</i>
pipes	Deprecated: <i>Removed in 3.13.</i>
pkgutil	<i>Utilities for the import system.</i>
platform	<i>Retrieves as much platform identifying data as possible.</i>
plistlib	<i>Generate and parse Apple plist files.</i>
poplib	<i>POP3 protocol client (requires sockets).</i>
posix	<i>The most common POSIX system calls (normally used via module os).</i>
pprint	<i>Data pretty printer.</i>
profile	<i>Python source profiler.</i>
pstats	<i>Statistics object for use with the profiler.</i>
pty	<i>Pseudo-Terminal Handling for Unix.</i>
pwd	<i>The password database (getpwnam() and friends).</i>
py_compile	<i>Generate byte-code files from Python source files.</i>
pyclbr	<i>Supports information extraction for a Python module browser.</i>
pydoc	<i>Documentation generator and online help system.</i>

q

queue	<i>A synchronized queue class.</i>
quopri	<i>Encode and decode files using the MIME quoted-printable encoding.</i>

r

random	<i>Generate pseudo-random numbers with various common distributions.</i>
re	<i>Regular expression operations.</i>
readline	<i>GNU readline support for Python.</i>
reprlib	<i>Alternate repr() implementation with size limits.</i>
resource	<i>An interface to provide resource usage information on the current process.</i>
rlcompleter	<i>Python identifier completion, suitable for the GNU readline library.</i>
runpy	<i>Locate and run Python modules without importing them first.</i>

s

sched	<i>General purpose event scheduler.</i>
secrets	<i>Generate secure random numbers for managing secrets.</i>
select	<i>Wait for I/O completion on multiple streams.</i>
selectors	<i>High-level I/O multiplexing.</i>
shelve	<i>Python object persistence.</i>
shlex	<i>Simple lexical analysis for Unix shell-like languages.</i>
shutil	<i>High-level file operations, including copying.</i>
signal	<i>Set handlers for asynchronous events.</i>

site	<i>Module responsible for site-specific configuration.</i>
sitecustomize	
smtpd	Deprecated: <i>Removed in 3.12.</i>
smtplib	<i>SMTP protocol client (requires sockets).</i>
sndhdr	Deprecated: <i>Removed in 3.13.</i>
socket	<i>Low-level networking interface.</i>
socketserver	<i>A framework for network servers.</i>
spwd	Deprecated: <i>Removed in 3.13.</i>
sqlite3	<i>A DB-API 2.0 implementation using SQLite 3.x.</i>
ssl	<i>TLS/SSL wrapper for socket objects</i>
stat	<i>Utilities for interpreting the results of <code>os.stat()</code>, <code>os.lstat()</code> and <code>os.fstat()</code>.</i>
statistics	<i>Mathematical statistics functions</i>
+ string	<i>Common string operations.</i>
stringprep	<i>String preparation, as per RFC 3453</i>
struct	<i>Interpret bytes as packed binary data.</i>
subprocess	<i>Subprocess management.</i>
sunau	Deprecated: <i>Removed in 3.13.</i>
symtable	<i>Interface to the compiler's internal symbol tables.</i>
+ sys	<i>Access system-specific parameters and functions.</i>
sysconfig	<i>Python's configuration information</i>
syslog	<i>An interface to the Unix syslog library routines.</i>
t	
tabnanny	<i>Tool for detecting white space related problems in Python source files in a directory tree.</i>
tarfile	<i>Read and write tar-format archive files.</i>
telnetlib	Deprecated: <i>Removed in 3.13.</i>
tempfile	<i>Generate temporary files and directories.</i>
termios	<i>POSIX style tty control.</i>
+ test	<i>Regression tests package containing the testing suite for Python.</i>
textwrap	<i>Text wrapping and filling</i>
threading	<i>Thread-based parallelism.</i>
time	<i>Time access and conversions.</i>
timeit	<i>Measure the execution time of small code snippets.</i>
+ tkinter	<i>Interface to Tcl/Tk for graphical user interfaces</i>
token	<i>Constants representing terminal nodes of the parse tree.</i>
tokenize	<i>Lexical scanner for Python source code.</i>
tomllib	<i>Parse TOML files.</i>
trace	<i>Trace or track Python statement execution.</i>
traceback	<i>Print or retrieve a stack traceback.</i>

tracemalloc	<i>Trace memory allocations.</i>
tty	<i>Utility functions that perform common terminal control operations.</i>
turtle	<i>An educational framework for simple graphics applications</i>
turtledemo	<i>A viewer for example turtle scripts</i>
types	<i>Names for built-in types.</i>
typing	<i>Support for type hints (see :pep:`484`).</i>
u	
unicodedata	<i>Access the Unicode Database.</i>
+ unittest	<i>Unit testing framework for Python.</i>
+ urllib	
usercustomize	
uu	Deprecated: <i>Removed in 3.13.</i>
uuid	<i>UUID objects (universally unique identifiers) according to RFC 9562</i>
v	
venv	<i>Creation of virtual environments.</i>
w	
warnings	<i>Issue warning messages and control their disposition.</i>
wave	<i>Provide an interface to the WAV sound format.</i>
weakref	<i>Support for weak references and weak dictionaries.</i>
webbrowser	<i>Easy-to-use controller for web browsers.</i>
winreg	<i>Routines and objects for manipulating the Windows registry.</i>
winsound	<i>Access to the sound-playing machinery for Windows.</i>
+ wsgiref	<i>WSGI Utilities and Reference Implementation.</i>
x	
xdrlib	Deprecated: <i>Removed in 3.13.</i>
+ xml	<i>Package containing XML processing modules</i>
+ xmlrpc	<i>Server and client modules implementing XML-RPC.</i>
z	
zipapp	<i>Manage executable Python zip archives</i>
zipfile	<i>Read and write ZIP-format archive files.</i>
zipimport	<i>Support for importing Python modules from ZIP archives.</i>
zlib	<i>Low-level interface to compression and decompression routines compatible with gzip.</i>
zoneinfo	<i>IANA time zone support</i>